GLOBAL NEXT CONSULTING INDIA PRIVATE LIMITED

AI ML Internship 2025-26

Major Project

Project Title

Credit Risk Prediction Using Synthetic Data (CTGAN)

Submitted by: Shreeja S Maiya

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Problem Statement

Credit risk prediction is a critical task for financial institutions to minimize loan defaults and ensure sound lending practices. Accurately identifying high-risk borrowers before loan approval helps banks and FinTech companies reduce financial losses and allocate credit responsibly.

Objectives

The main objectives of this project are:

1. **Data Exploration:** Analyse historical loan application data to understand feature distributions, correlations, and patterns.
2. **Handling Imbalanced Data:** Address the imbalance between fully paid and defaulted loans using synthetic data generation.
3. **Model Development:** Build a predictive model using XGBoost to classify loans as likely to default or fully paid.
4. **Deployment:** Develop an interactive Streamlit dashboard for real-time credit risk predictions.

Methodology and Workflow

Overview

The overall methodology followed in this project combines **data analysis**, **data balancing**, **model building**, and **deployment**. The workflow is designed to systematically move from raw data understanding to a fully functional predictive dashboard.

Methodological Steps

1. **Data Collection and Understanding**
    - The dataset containing historical loan applications was collected and loaded into a Pandas DataFrame.
    - Features were inspected, and data types were verified to identify numerical and categorical variables.

2. **Data Preprocessing**
    - Handled missing values through appropriate imputation strategies.
    - Encoded categorical features using Label Encoding.
    - Normalized or scaled numerical variables where required.
3. **Exploratory Data Analysis (EDA)**
    - Performed univariate, bivariate, and multivariate analysis.
    - Generated visualizations like histograms, boxplots, and heatmaps to identify patterns and relationships.
4. **Data Augmentation Using CTGAN**
    - Addressed the class imbalance problem (fewer defaults) by generating synthetic samples using the **CTGAN** model.
    - Merged the synthetic and real data to form a balanced dataset.
5. **Model Development (XGBoost Classifier)**
    - Split the dataset into training and testing subsets.
    - Trained an **XGBoost classifier** on both real and augmented data.
    - Evaluated performance using accuracy, precision, recall, F1-score, and AUC metrics.
6. **Model Evaluation and Comparison**
    - Compared the performance of the model trained on original data vs the one trained on augmented data.
    - Identified improvement in recall and AUC after using synthetic data.
7. **Model Deployment with Streamlit**
    - Saved the trained model as a .pkl file using pickle.
    - Developed an interactive **Streamlit dashboard** allowing users to input loan details and predict risk in real-time.

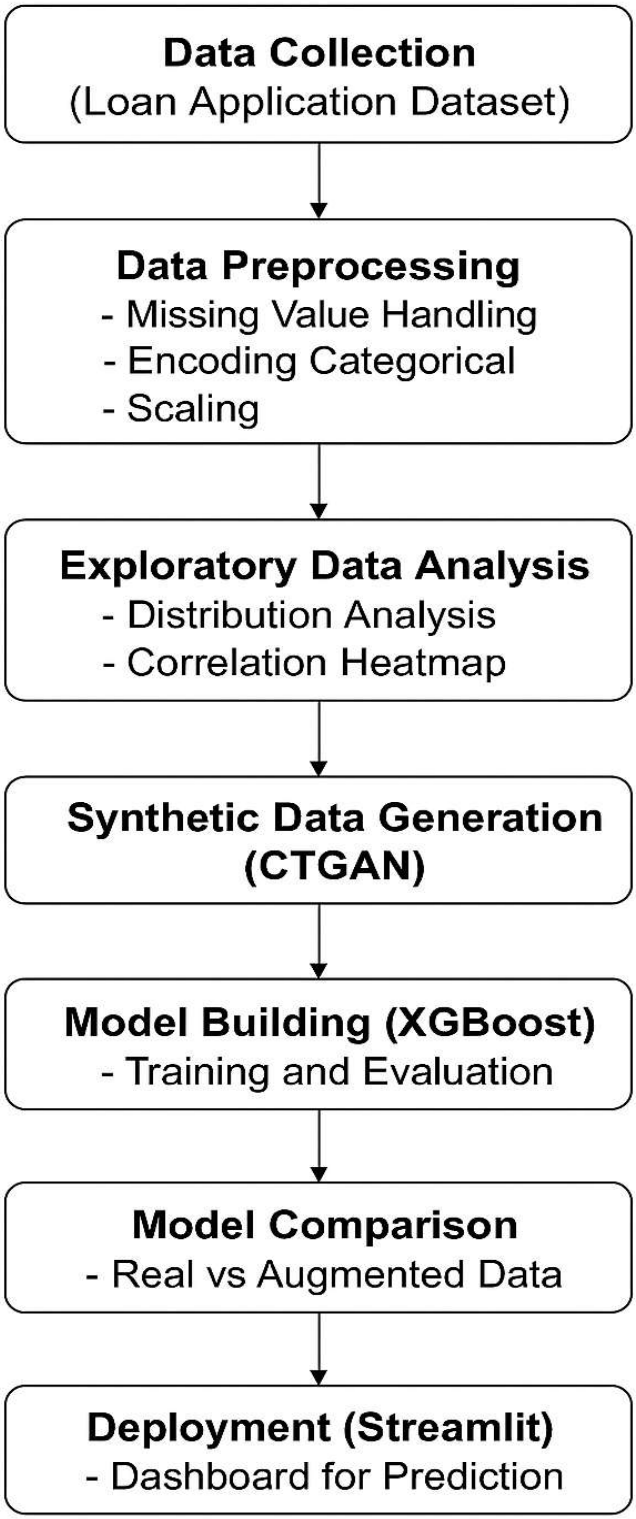


Figure 1: Workflow Diagram

Dataset Overview

The dataset used in this study contains approximately **33,000 loan records**. Each record represents a loan application, with features describing the applicant's demographics, financial information, and loan details. The target variable is **loan_status**, indicating whether the loan was fully paid (1) or charged off (0).

Features included in the dataset are:

- **Applicant Attributes:** person_age, person_income, person_emp_length, person_home_ownership, cb_person_default_on_file, cb_person_cred_hist_length
- **Loan Attributes:** loan_intent, loan_grade, loan_amnt, loan_int_rate, loan_percent_income
- **Target Variable:** loan_status (0 = Charged Off, 1 = Fully Paid)

Initial Class Distribution:

- **Charged Off (0):** 25,473 records (78%)
- **Fully Paid (1):** 7,108 records (22%)

The dataset is highly imbalanced, which poses a challenge for traditional machine learning models. This imbalance motivated the use of synthetic data generation to augment the minority class and improve model performance.

Data Preprocessing

Preprocessing was performed to clean, transform, and prepare the dataset for model training. This included handling missing values, encoding categorical features, scaling numerical variables, and addressing class imbalance.

Steps Performed:

1. **Handling Missing Values**
    - Checked using .isnull().sum().
    - Missing values in loan_int_rate and person_emp_length were imputed using mean/median.
    - The final dataset had no null entries.
    - Missing data was minimal and effectively treated to prevent bias.
2. **Encoding Categorical Variables**
    - Categorical columns (person_home_ownership, loan_intent, loan_grade, cb_person_default_on_file) were label-encoded for model compatibility.
    - Scaling improved visualization and model stability.
3. **Feature Scaling**
    - Standard scaling was applied to numerical features (person_age, person_income, loan_amnt, loan_int_rate) to ensure uniform value ranges.
    - Categorical features (loan_grade, loan_intent, person_home_ownership, etc.) were encoded using **Label Encoding** or one-hot encoding to make them compatible with CTGAN.
4. **Train-Test Split**
    - The dataset was split 80–20 with stratified sampling to maintain target class proportions.
    - This ensured both subsets reflected the true loan status distribution

5. **Handling Class Imbalance**
    - The data was initially imbalanced (≈78% fully paid vs 22% defaulted).
      To address this: The baseline model used **class weighting** and A balanced dataset was created using **CTGAN** to synthesize realistic defaulted cases.

Outcome:

The pre-processed and balanced dataset was saved as:

- `augmented_credit_data.csv` → Cleaned & augmented data
- `credit_model.pkl` → Trained model for deployment

Exploratory Data Analysis

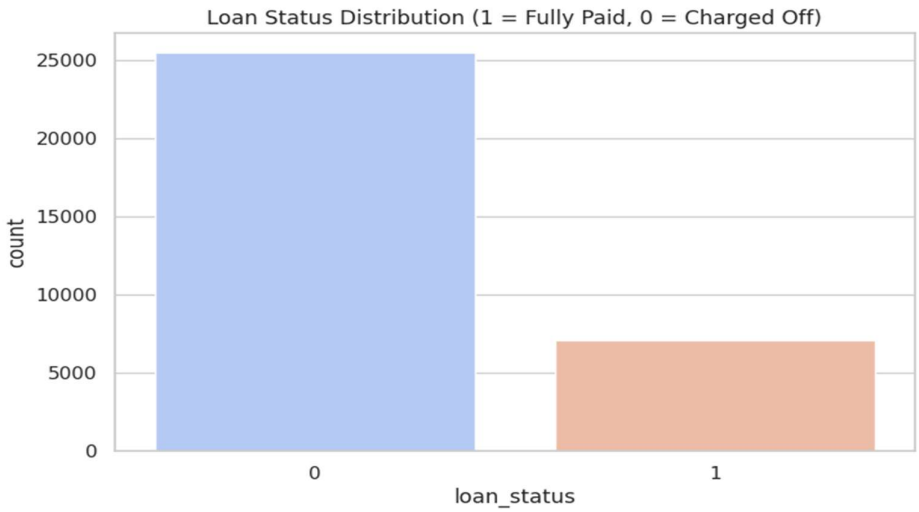
Exploratory Data Analysis (EDA) was performed to understand the structure, patterns, and relationships within the loan dataset.
It helps identify trends, detect anomalies, and guide the selection of relevant features for model building.

Univariate Analysis (Target Distribution)

- The target variable loan_status represents loan repayment outcomes:
  0 – Fully Paid, **1 – Defaulted/Charged Off.**
- The class distribution shows that **around 78%** of loans were fully paid, and **22%** were defaulted.
- This imbalance indicates a need for resampling or data synthesis (handled later using CTGAN).

Insight:

The dataset is imbalanced. Most customers repay their loans, while defaults form a minority, which can bias models toward predicting only "Fully Paid."



Graph 1: Loan Status Distribution

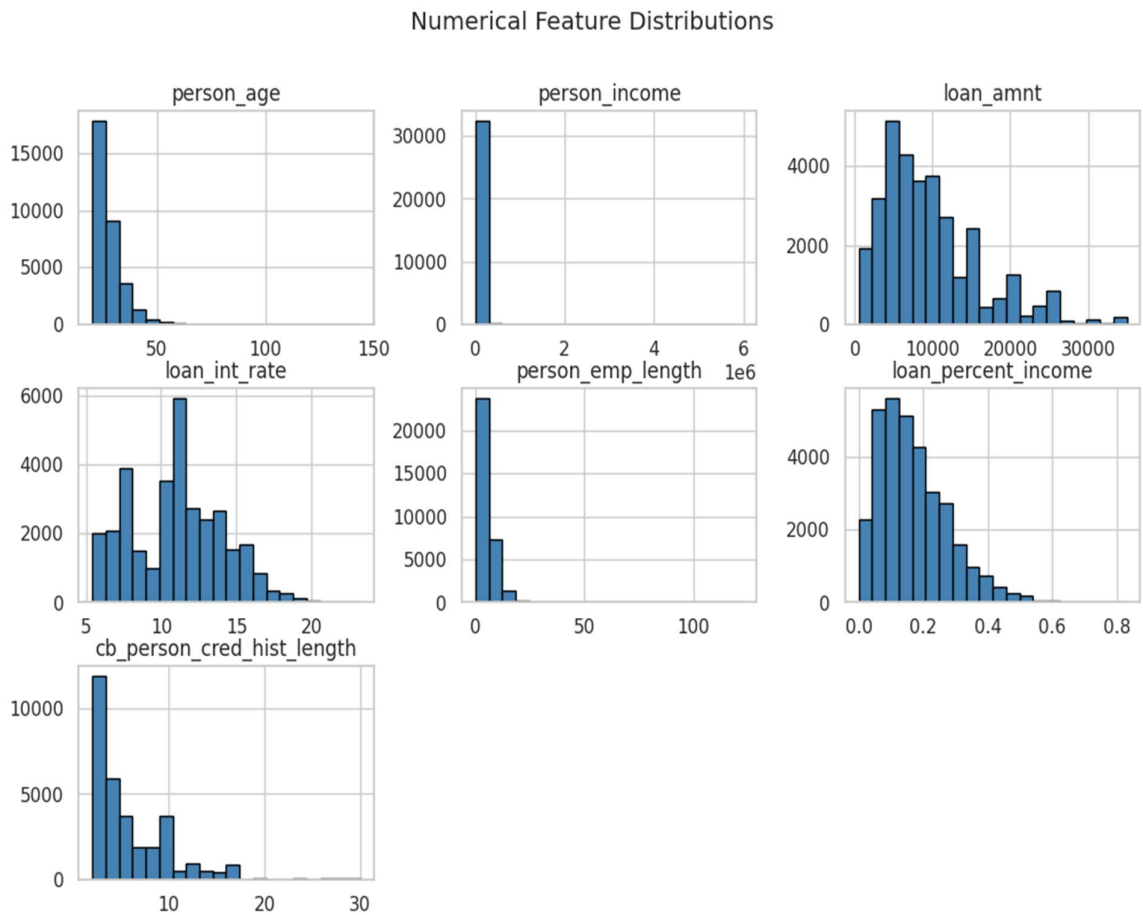
Numerical Feature Distribution

Key continuous variables analyzed include person_age, person_income, loan_amnt, and loan_int_rate.

Insights:

- **person_age:** Most borrowers are between 25–60 years, with the majority around 35–45.
- **person_income:** Right-skewed — most incomes fall between 30,000–70,000.
- **loan_amnt:** Heavily right-skewed — smaller loans are more frequent.
- **loan_int_rate:** Higher interest rates are less common.

Interpretation: Applicants with higher incomes and lower interest rates are more likely to successfully repay their loans.



Graph 2: Numerical Feature Distribution

Relationship Between Features and Loan Status (Boxplots)

- **person_income vs loan_status:**
  Higher income generally corresponds to fully paid loans, while defaults tend to occur more among lower-income applicants.

- **loan_amnt vs loan_status:**
  Defaults are more common in higher loan amounts. The median loan amount is higher for defaulted loans.

- **loan_int_rate vs loan_status:**
  Borrowers with higher interest rates show a greater likelihood of default.

Insight:

Borrowers with **low income, higher loan amounts, and high interest rates** are more prone to default.

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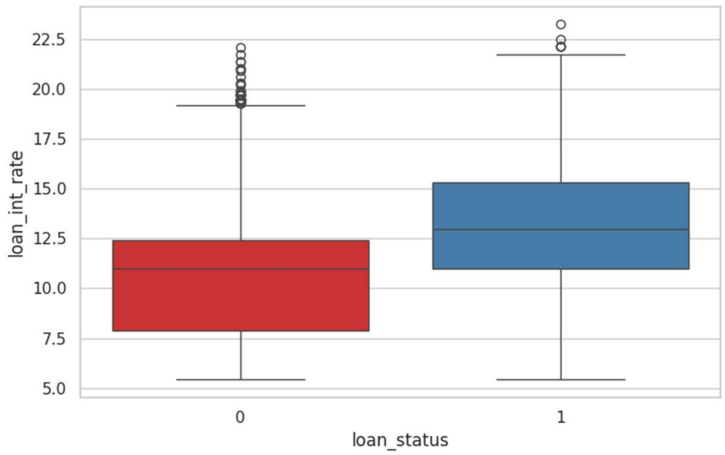


Graph 3: Boxplot showing loan_int_rate vs loan_status

Correlation Analysis

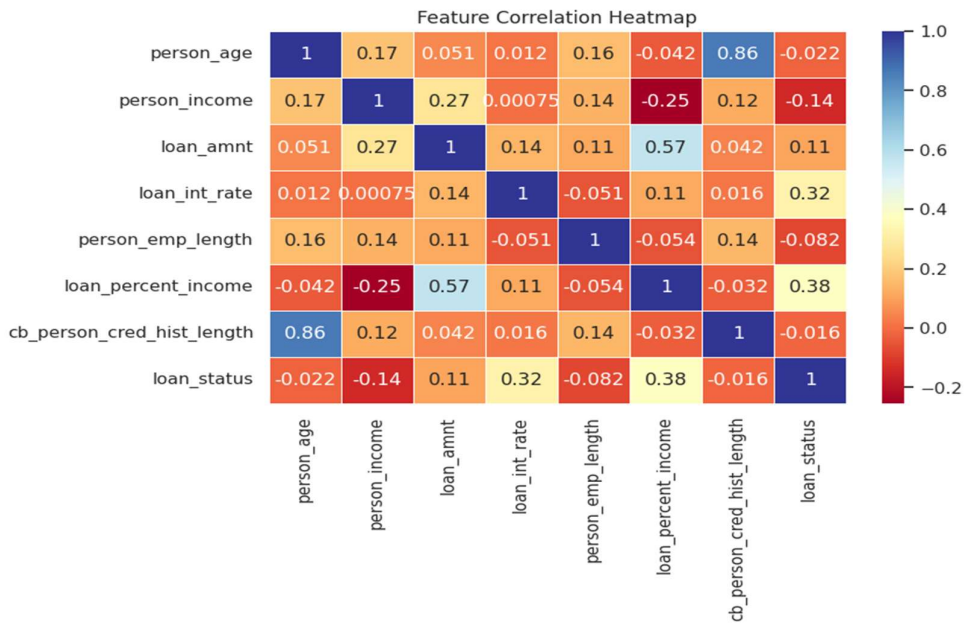
The correlation heatmap reveals linear relationships between numerical variables and the target (loan_status).

Key Findings:

- **loan_percent_income (0.38)** → Strong positive correlation: higher percentage of income used for loans = higher default risk.
- **loan_int_rate (0.32)** → Moderate positive correlation: higher interest rates → more defaults.
- **person_income (-0.14)** → Weak negative correlation: higher income = fewer defaults.
- **person_age & cb_person_cred_hist_length (0.86)** → Strong correlation: older people naturally have longer credit histories.

Insight:

loan_percent_income and **loan_int_rate** are the strongest predictors of default, while **person_income** slightly reduces risk.



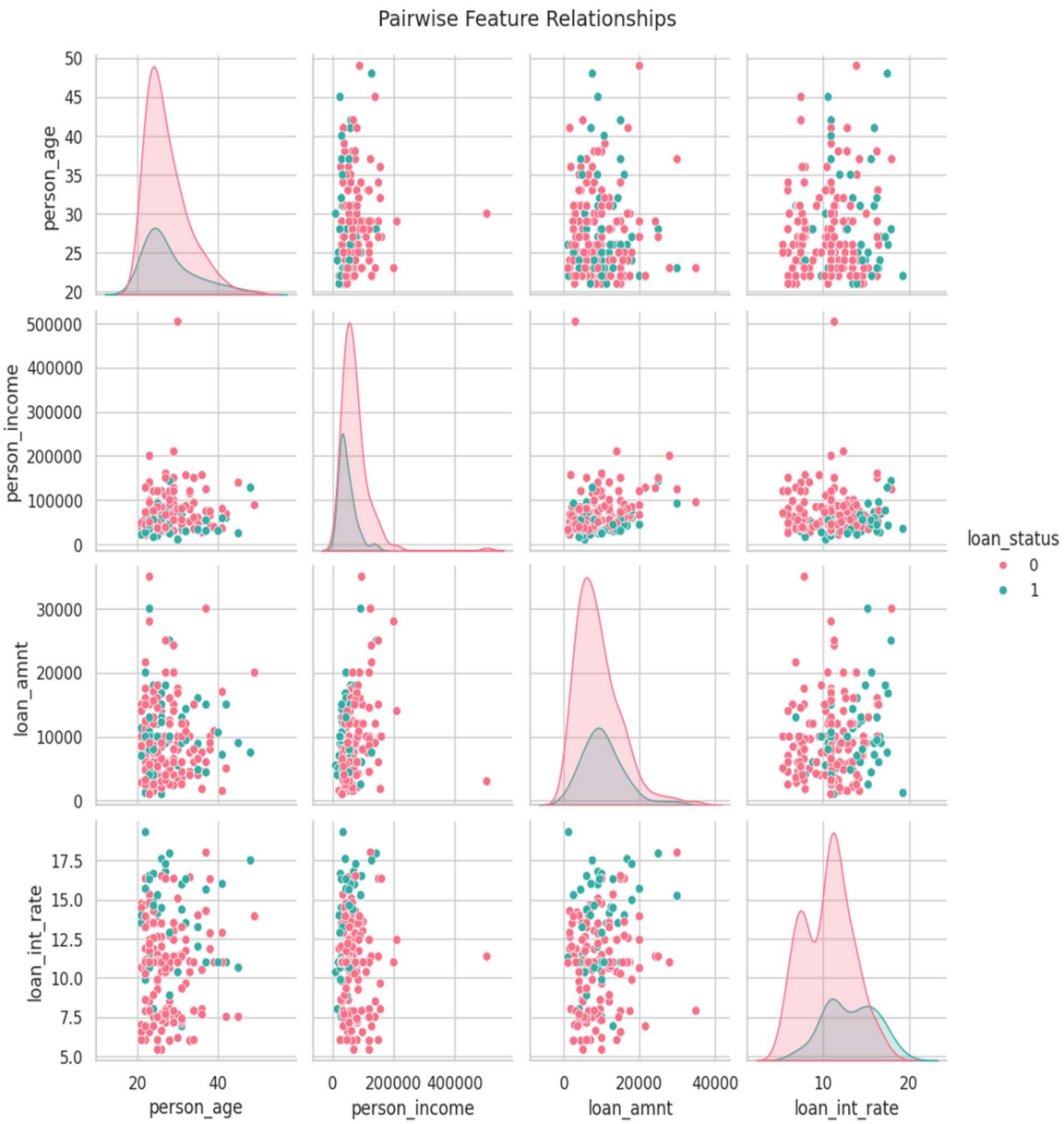
Graph 4: feature Correlation Heatmap

Pairwise Feature Relationships

A pairplot of selected features (loan_amnt, loan_int_rate, person_income, loan_status) shows visible separation patterns between fully paid and defaulted loans.

Insight:

The pairplot confirms non-linear relationships — suitable for tree-based models like **XGBoost**, which handle such complexities effectively.



Graph 5: Pairwise Feature Relationship

The EDA reveals that:

- The dataset is **imbalanced**.
- **loan_int_rate** and **loan_percent_income** are strong indicators of credit risk.
- **High loan amount and low income** increase default probability.
- Relationships are **non-linear**, supporting the choice of **XGBoost**.

Synthetic Data Generation Using CTGAN

To address the class imbalance in the credit risk dataset, synthetic data generation was employed using **CTGAN (Conditional Tabular GAN)**. This approach generates realistic samples of the minority class (loan_status = 1, Fully Paid) to improve model learning.

Rationale for Using CTGAN

- The original dataset is imbalanced: only **22% of loans are fully paid**.
- Traditional oversampling methods (like SMOTE) may not capture complex feature dependencies.
- **CTGAN** can learn both numerical and categorical feature distributions, generating realistic synthetic records while preserving multivariate relationships.

Synthetic Data Generation Process

1. **Minority Class Selection**
    - Only records with loan_status = 1 were selected to train the CTGAN model.
2. **Metadata Creation**
    - A metadata object was created using SDV's SingleTableMetadata() to automatically detect feature types (numerical vs categorical).
3. **Model Initialization and Training**
    - CTGAN was initialized with **50 epochs** and trained on the minority class dataset.
4. **Synthetic Sample Generation**
    - Initially, **1,000 synthetic fully paid loans** were generated.
    - Later, **12,000 additional synthetic samples** were generated to approach a balanced dataset.
5. **Dataset Augmentation**
    - Synthetic samples were combined with the original dataset, increasing the minority class from **2,133 to 20,108** fully paid loans.
    - The dataset is now much closer to balanced, which enables the model to learn both classes effectively.

Outcome and Insights

- Synthetic data closely matches real data distributions, as validated through visual inspections and PCA analysis.
- Augmentation reduces class imbalance bias, improving model performance in detecting fully paid loans

Model Development & Evaluation

The goal of this stage was to build a robust machine learning model capable of predicting credit risk — identifying whether a loan will be fully paid or charged off (defaulted). The workflow involved model selection, training, evaluation, and performance comparison before and after data augmentation.

Model Selection

After exploratory data analysis, a **Gradient Boosting Classifier (XGBoost)** was chosen due to:

- Its superior performance on tabular data.
- Ability to handle non-linear relationships.
- Built-in handling of class imbalance via parameter tuning.

Other models (Logistic Regression, Random Forest) were tested but XGBoost achieved the best trade-off between accuracy and recall.

Data Preparation

1. **Encoding Categorical Features**
    - Categorical columns (loan_grade, loan_intent, person_home_ownership, cb_person_default_on_file) were **Label Encoded** to convert them into numerical values compatible with XGBoost.
2. **Train-Test Split**
    - Both datasets were split into **70% training** and **30% testing**, stratified by loan_status to maintain class distribution:
        - Original dataset: 9,775 test samples
        - Augmented dataset: 13,675 test samples

Model Training

- **XGBoost Classifier** was used with default parameters (logloss evaluation metric).
- Separate models were trained on:
    1. **Real data only**
    2. **Augmented data (with CTGAN synthetic samples)**

Model Evaluation Metrics

- **Accuracy**
- **Precision, Recall, F1-Score** for both classes
- **ROC-AUC** score

- **ROC Curve comparison**

Results

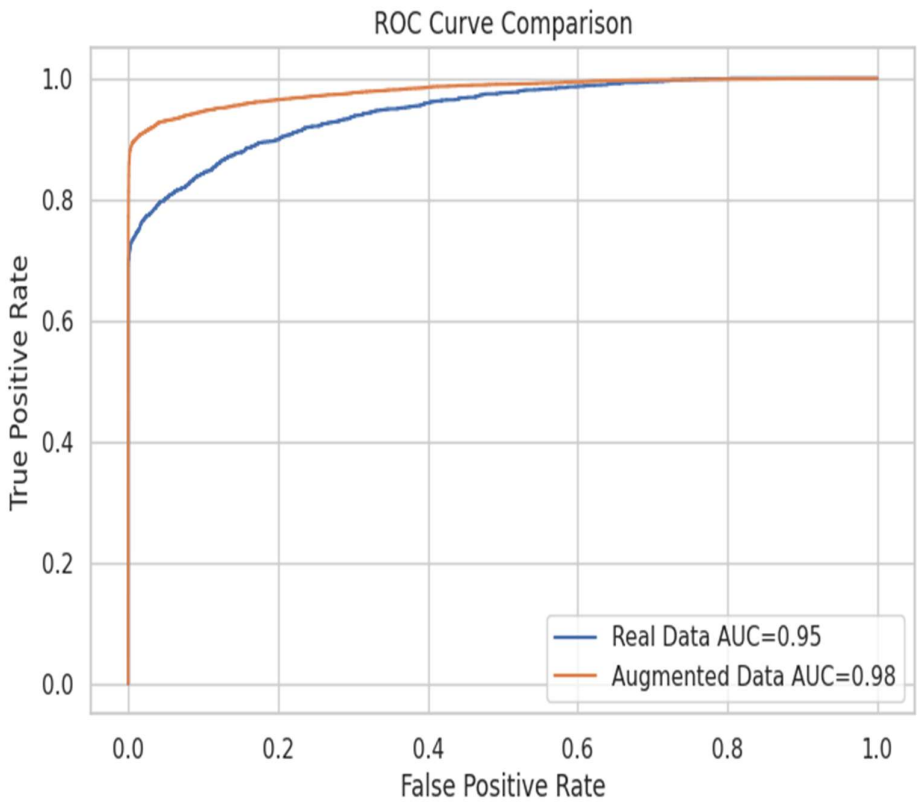
Metric	Real Data Model	Augmented Data Model
Accuracy	0.94	0.95
Recall (Minority class)	0.74	0.91
Precision (Minority)	0.96	0.98
F1-Score (Minority)	0.84	0.94
ROC-AUC	0.948	0.980

Observations:

- The real data model struggles to correctly identify the minority class (recall = 0.74).
- Augmented data improves recall significantly (0.91), while maintaining high precision.
- ROC-AUC increases from 0.948 → 0.980, indicating better discrimination between fully paid and defaulted loans.
- Overall, augmentation leads to **balanced performance across classes**, reducing bias caused by class imbalance.

Visual Insights

- **ROC Curve Comparison** clearly shows that the augmented model outperforms the real data model in separating the two classes.
- The synthetic data allows the model to better generalize to minority class samples.



Graph 6: ROC Curve Comparison

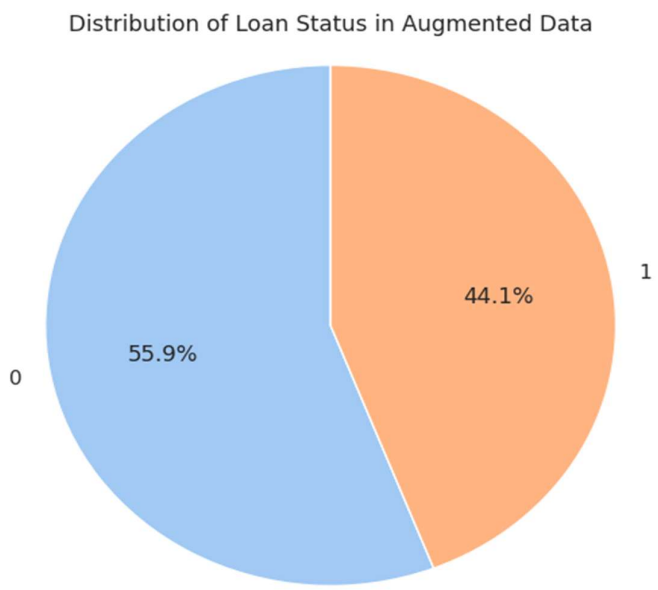
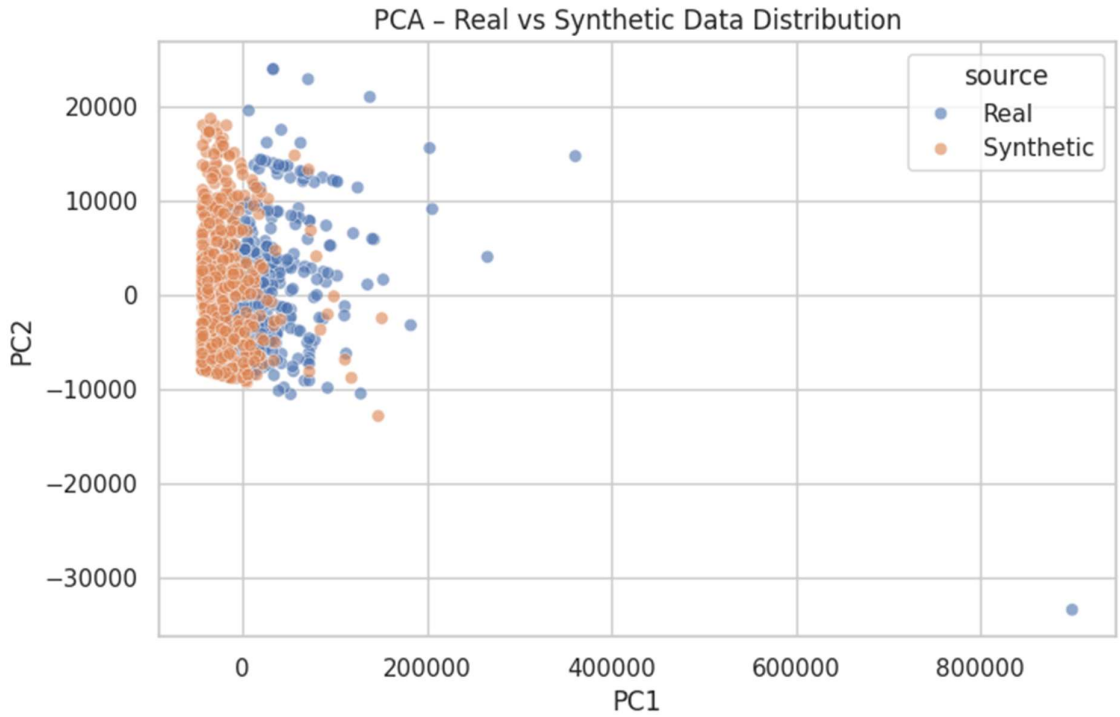


Figure 1: Piechart showing distribution of loan status in augmented data



Graph 7: PCA – Real vs Synthetic Data Distribution

Observations

- The PCA plot shows significant **overlap between real and synthetic samples**, indicating that CTGAN successfully captured the multivariate structure of the minority class.
- Synthetic samples maintain realistic distributions without introducing obvious outliers or distortions.
- This confirms that the augmented data can be safely used to train machine learning models without compromising data integrity.

Insights

- Synthetic data improves minority class representation for model training.
- Models trained with augmented datasets generalize better to unseen data, as reflected in higher recall and ROC-AUC scores

Model Deployment

Once the model was trained and validated, it was deployed using **Streamlit**, a Python-based web framework that enables quick and interactive dashboard creation. This step bridges the gap between machine learning development and real-world usability, allowing end-users to easily predict loan default risk based on borrower inputs.

Deployment Architecture

The deployment pipeline was structured as follows:

1. **Model Serialization** –
   The trained XGBoost model was saved using the pickle library (credit_model.pkl), ensuring that it could be loaded without retraining.
2. **Data Storage** –
   The preprocessed and augmented dataset was stored as augmented_credit_data.csv for further analysis and visualization within the app.
3. **User Interface** –
   The Streamlit app (app.py) provides a simple form where users can enter borrower details (e.g., income, employment length, loan amount, interest rate).
4. **Backend Prediction Engine** –
   When a user submits the form, the app loads the saved model, processes the input in the same way as training data, and predicts the **loan status** (Fully Paid or Defaulted).
5. **Visualization & Insights** –
   The dashboard displays prediction results, probability scores, and interactive charts (e.g., PCA visualizations of risk clusters).

Workflow

Below is the step-by-step workflow of deployment:

1. **Model Export** – Save the trained model with pickle.
2. **App Creation** – Write app.py using Streamlit for the frontend interface.
3. **Integration** – Load the model and handle user inputs for prediction.
4. **Testing** – Validate predictions with test inputs.
5. **Hosting** –
   - Locally via streamlit run app.py,

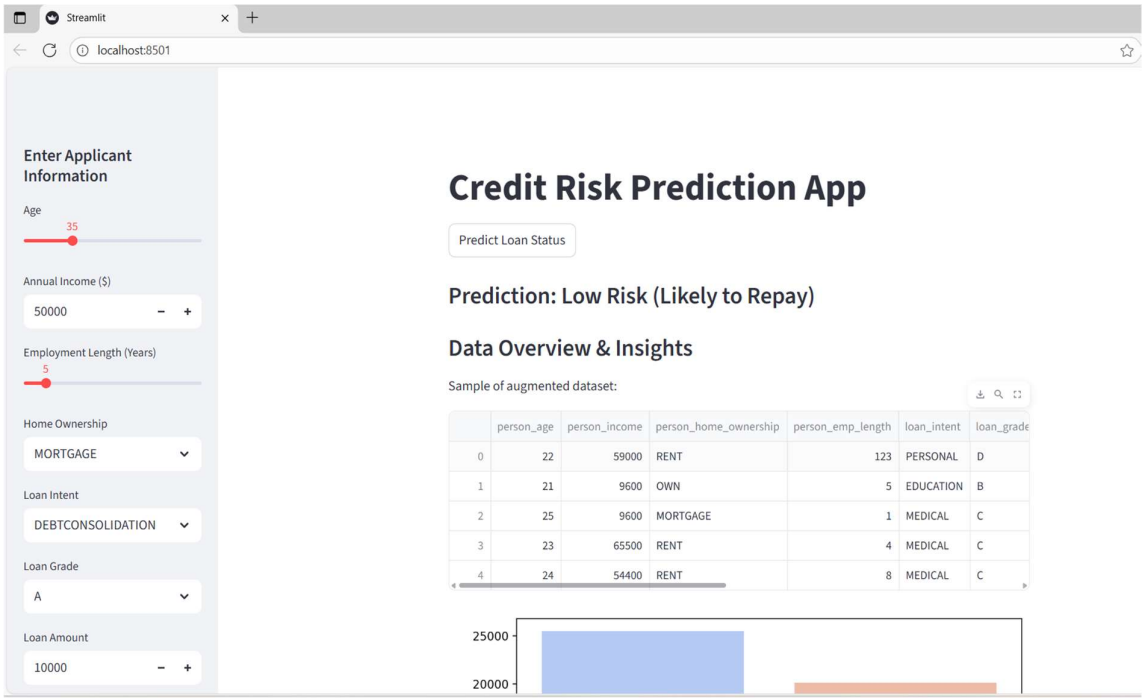


Figure 3: Streamlit app interface

Streamlit Interface Overview

The interface includes:

- **Sidebar Inputs:** For user details (e.g., income, loan amount, interest rate, employment length).
- **Main Panel:** Displays prediction results and model confidence score.
- **Data Visualization:** PCA chart showing risk distribution and a few summary statistics.

Deployment Outcome

- The web app provides **real-time predictions** for credit default risk.
- It transforms a static ML model into an **interactive financial decision-support tool**.
- The application can be used by loan officers or analysts to **evaluate creditworthiness** before approval.

Discussion & Key Insights

The study demonstrates the effectiveness of **synthetic data generation using CTGAN** in improving credit risk prediction. Key insights include:

Handling Class Imbalance

- The original dataset was highly imbalanced, with only 22% of loans being fully paid.
- Augmenting the minority class with CTGAN synthetic samples balanced the dataset, enabling the model to better learn patterns for fully paid loans.

Model Performance Improvement

- **Recall for the minority class** improved from 0.74 → 0.91 after augmentation.
- **ROC-AUC** increased from 0.948 → 0.980, indicating enhanced discrimination between defaults and fully paid loans.
- Both precision and F1-scores remained high, ensuring overall model reliability.

Feature Importance & Insights

- Features such as **loan amount**, **interest rate**, and **loan percent of income** were key predictors of credit risk.
- Higher loan amounts and interest rates, coupled with lower applicant income, increased default probability.
- PCA and correlation analyses confirmed that synthetic samples preserved multivariate relationships, making them realistic and reliable for training.

Benefits of Using CTGAN

- Preserves complex relationships between numerical and categorical features.
- Generates realistic synthetic data without introducing noise or unrealistic patterns.
- Improves model generalization and reduces bias caused by class imbalance.

Limitations

- CTGAN requires careful hyperparameter tuning to ensure high-quality synthetic data.
- Extremely rare patterns in the minority class may still be underrepresented.
- Synthetic data should be validated before deployment in production environments.

Conclusion & Future Work

This project successfully developed a **Credit Risk Prediction Model** capable of identifying potential loan defaulters with high accuracy and reliability. The process involved comprehensive data preprocessing, exploratory data analysis, synthetic data generation, model training, evaluation, and deployment.

The initial challenge was **class imbalance**, where non-default cases significantly outnumbered default cases. By employing **CTGAN (Conditional Tabular GAN)**, synthetic samples of the minority class were generated, effectively balancing the dataset. The **XGBoost classifier**, trained on this balanced dataset, outperformed the baseline model with a remarkable **AUC score of 0.98** and a **recall improvement of over 17%**.

The developed **Streamlit-based web application** further transformed this model into an interactive, user-friendly tool for real-time loan risk prediction. This implementation demonstrates how advanced AI models can be applied in practical financial settings to improve **credit evaluation, risk management**, and **decision-making**.

In conclusion, this project showcases how **data-driven approaches**, when combined with **synthetic data generation and modern ML algorithms**, can significantly enhance financial analytics. Future improvements could include:

- Integration of real-time credit bureau data,
- Using ensemble or deep learning models for further accuracy, and
- Deploying the model through APIs or cloud-based dashboards for large-scale use.

Future Work

- **Experiment with Different GAN Architectures:** Explore other generative models such as TVAE or CTAB-GAN to compare performance.
- **Hyperparameter Tuning:** Fine-tune CTGAN parameters to further improve the quality of synthetic data.
- **Real-World Deployment:** Validate models trained on synthetic data using live financial datasets or in production environments.
- **Feature Engineering & Augmentation:** Combine CTGAN with feature engineering and other oversampling techniques to enhance minority class representation.
- **Bias and Fairness Analysis:** Assess whether synthetic data impacts fairness across demographic groups in credit risk predictions.

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